

Final Assignment

Task 1: Variable Classification & Coding (10 Marks)

Identify:

- Nominal variables
- Ordinal variables
- Scale variables

Then perform:

👉 Recode:

Screen_Time_Group

- ≤ 5 Hours $\rightarrow$ Low
- 6–8 Hours $\rightarrow$ Moderate
- 8 Hours $\rightarrow$ High

👉 Create:

Expense_Group

- $< 6000 \rightarrow$ Low Expense
- 6000–10000 $\rightarrow$ Medium Expense
- 10000 $\rightarrow$ High Expense

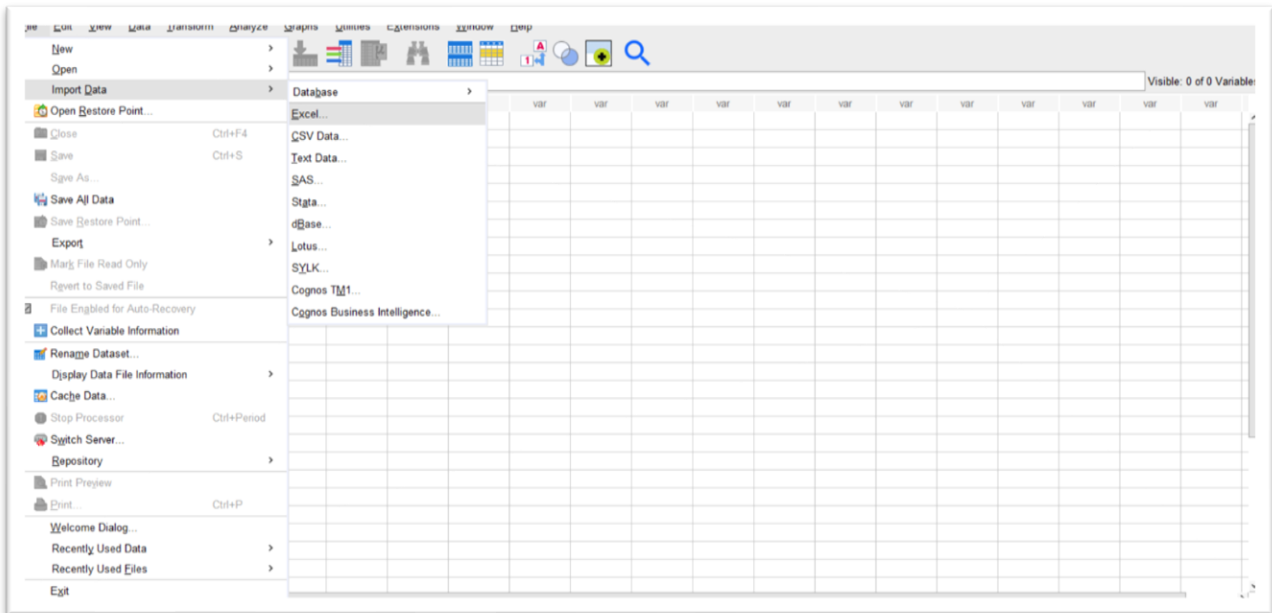
👉 Create:

Productivity_Level

- Productivity Score $\geq 70 \rightarrow$ High
- Productivity Score $< 70 \rightarrow$ Low

📌 Interpretation Not Required.

First, we need to import the dataset into SPSS. Importing data from different sources in SPSS is very simple. Go to the **File** tab $\rightarrow$ **Import Data** (or **Open** $\rightarrow$ **Data**) $\rightarrow$ then select your source file type (Excel, CSV, etc.) and choose the dataset file.



	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	Participant_ID	String	4	0		None	None	4	Left	Nominal	Input
2	Transport_Mode	String	8	0		None	None	8	Left	Nominal	Input
3	Daily_Screen_Time_Hours	Numeric	2	0		None	None	12	Right	Scale	Input
4	Exercise_Days_Per_Week	Numeric	1	0		None	None	12	Right	Scale	Input
5	Food_Habit	String	9	0		None	None	9	Left	Nominal	Input
6	Coffee_Consumption	String	8	0		None	None	8	Left	Nominal	Input
7	Monthly_Expense_BDT	Numeric	5	0		None	None	12	Right	Scale	Input
8	Attendance_Percentage	Numeric	2	0		None	None	12	Right	Scale	Input
9	Exercise_Level	String	6	0		None	None	6	Left	Nominal	Input
10	Productivity_Score	Numeric	4	1		None	None	12	Right	Scale	Input
11	Screen_Time_Group	Numeric	8	0	{1, Low}...	None	None	19	Right	Ordinal	Input
12	Expense_Group	Numeric	8	0	{1, Low Exp...	None	None	15	Right	Ordinal	Input
13	Productivity_Level	Numeric	8	0	{1, Low}...	None	None	20	Right	Ordinal	Input
14											
15											
16											
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28											
29											

✓ Variable Classification

◆ Nominal Variables

- Participant_ID
- Transport_Mode
- Food_Habit
- Coffee_Consumption

◆ **Ordinal Variables**

- Exercise_Level
- Screen_Time_Group
- Expense_Group
- Productivity_Level

◆ **Scale Variables**

- Daily_Screen_Time_Hours
- Exercise_Days_Per_Week
- Monthly_Expense_BDT
- Attendance_Percentage
- Productivity_Score

Task 2: Descriptive Statistics & Explore (10 Marks)

Generate:

- Mean
- Median
- Standard Deviation
- Skewness
- Kurtosis

For:

- Productivity Score
- Daily Screen Time

- Monthly Expense

Run **the explore function** for:

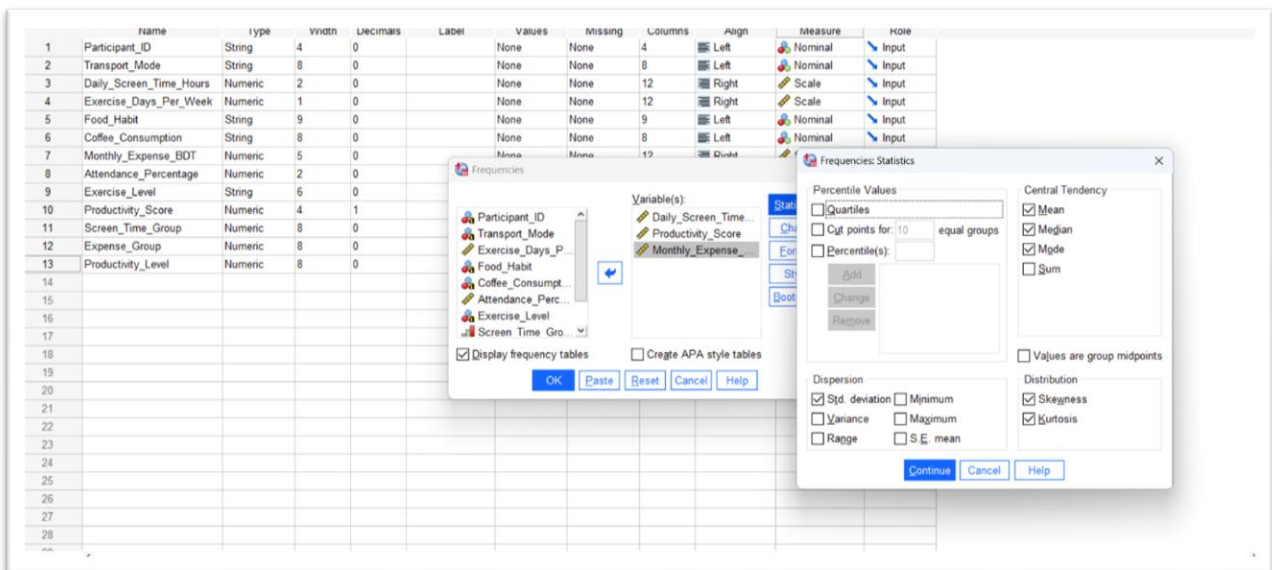
Productivity Score

Check:

- Normality
- Outliers

✦ Interpretation Required:

- What do the mean and standard deviation indicate?
- Is the data normally distributed?
- Are there any outliers?
- What do skewness and kurtosis suggest about the data?



Statistics

	Daily_Screen_Time_Hours	Productivity_Score	Monthly_Expense_BDT

N	Valid	100	100	100
	Missing	0	0	0
Mean		7.46	59.818	9295.62
Median		8.00	59.350	9442.50
Mode		4	40.0 ^a	11472
Std. Deviation		2.938	13.2903	3316.805
Skewness		-.075	.110	-.182
Std. Error of Skewness		.241	.241	.241
Kurtosis		-1.358	-.517	-.875
Std. Error of Kurtosis		.478	.478	.478
a. Multiple modes exist. The smallest value is shown				

Daily Screen Time Hours

The mean daily screen time is **7.46 hours**, which indicates that participants spend around 7 to 8 hours on screen daily.

The standard deviation is **2.938**, suggesting a moderate variation in screen time among participants.

The skewness value (-0.075) is close to zero, indicating that the data is approximately symmetric.

The kurtosis value (-1.358) indicates a relatively flatter distribution than normal.

Productivity Score

The mean productivity score is **59.82**, which represents the average productivity level of participants.

The standard deviation is **13.29**, indicating moderate variability in productivity scores.

The skewness value (0.110) is close to zero, suggesting the distribution is approximately normal and slightly positively skewed.

The kurtosis value (-0.517) indicates a slightly flatter distribution compared to a normal distribution.

Monthly Expense (BDT)

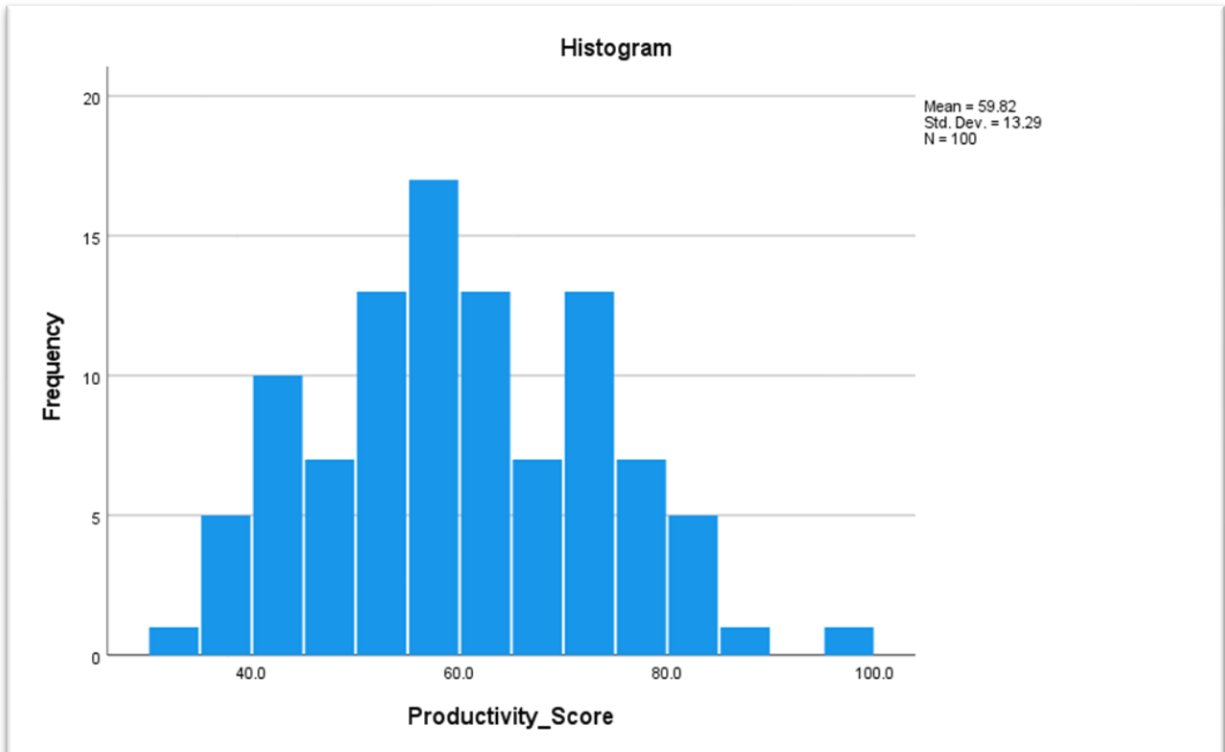
The average monthly expense of participants is **9295.62 BDT**.

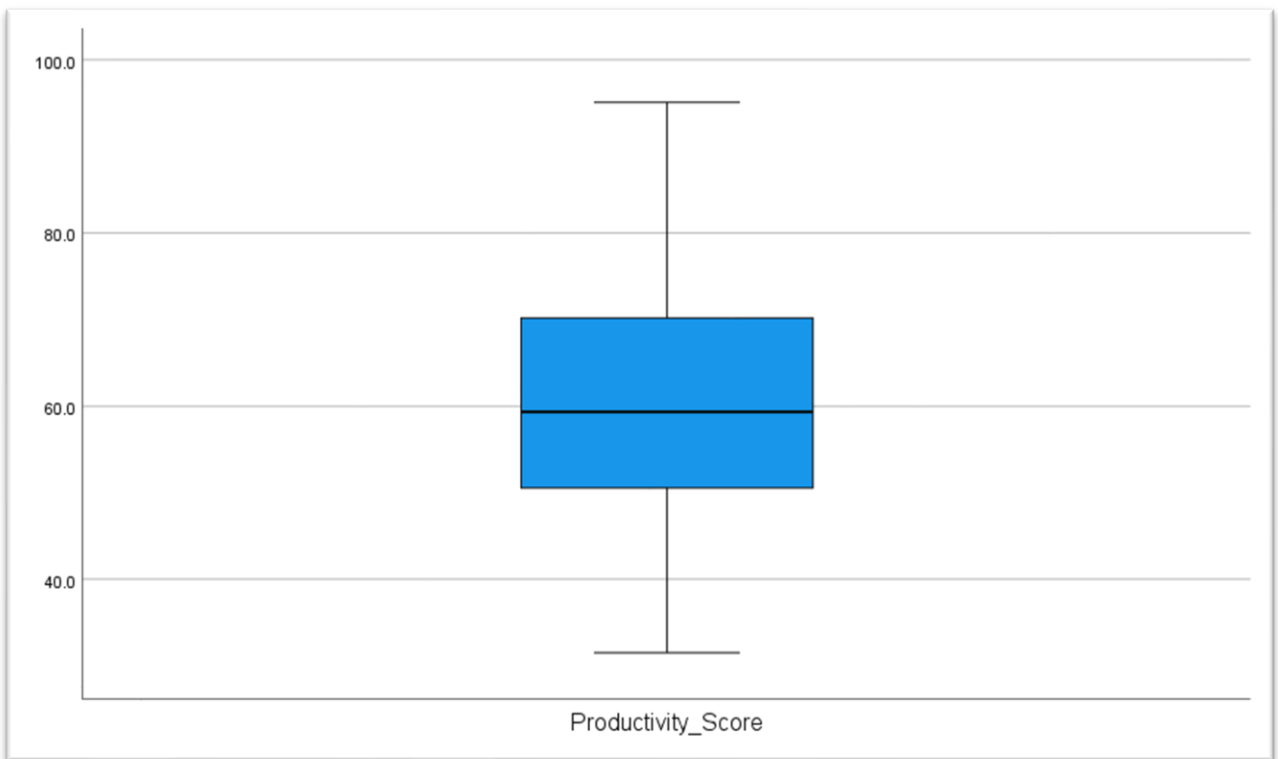
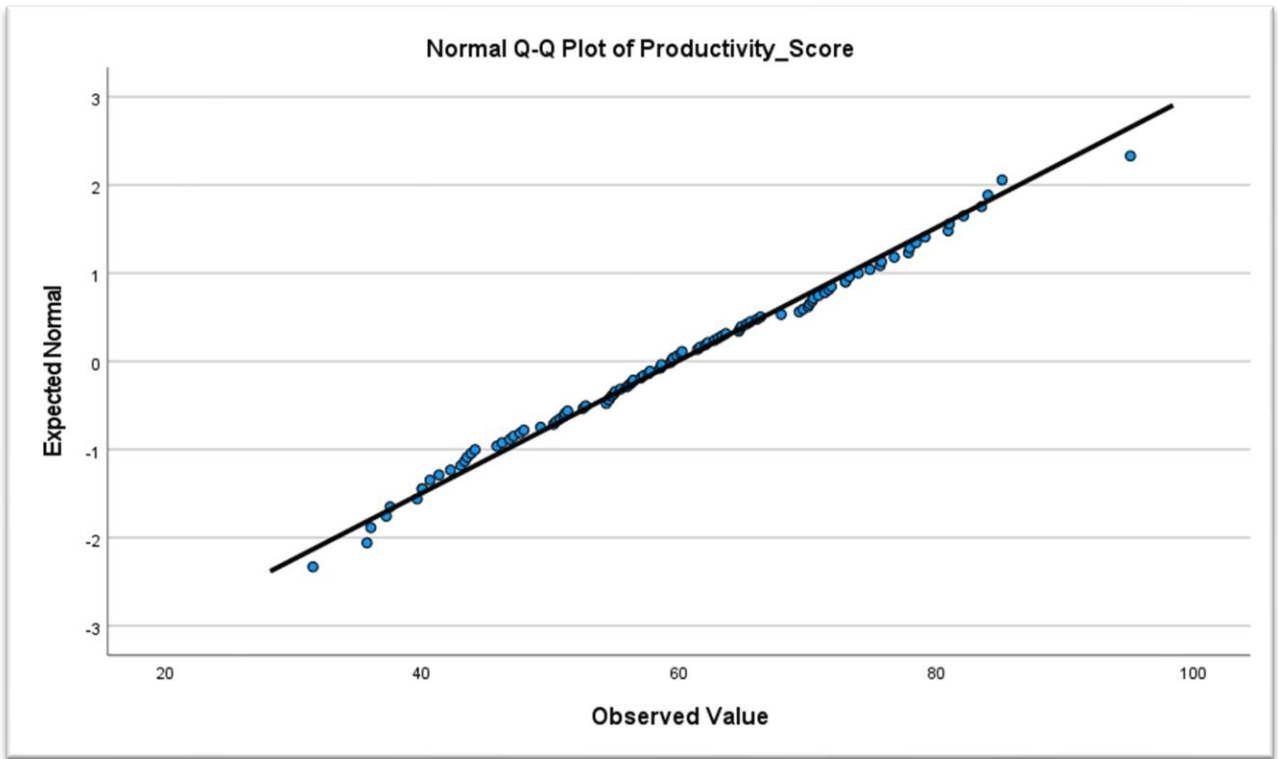
The standard deviation is **3316.81**, which suggests that participants' expenses vary considerably.

The skewness value (-0.182) indicates the data is slightly negatively skewed but still close to symmetric.

The kurtosis value (-0.875) suggests a flatter distribution than normal.

Tests of Normality						
	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Productivity_Score	.052	100	.200*	.990	100	.663
*. This is a lower bound of the true significance.						
a. Lilliefors Significance Correction						





Normality Test

The Shapiro-Wilk test shows a significance value of **0.663**, which is greater than **0.05**. Therefore, the productivity score data is normally distributed.

Histogram Interpretation

The histogram shows that the data is approximately bell-shaped and symmetric. Most participants' productivity scores are concentrated around the middle values, indicating a normal distribution pattern.

Q-Q Plot Interpretation

In the Normal Q-Q Plot, most data points lie close to the diagonal line. This suggests that the productivity score follows a normal distribution with only slight deviations.

Outlier Interpretation

The boxplot does not show any significant outliers because there are no extreme points or circles outside the whiskers. Therefore, the dataset appears free from major outliers.

Task 3: Cross-tabulation & Association (10 Marks)

Run:

Screen_Time_Group × Productivity_Level

Include:

- Row %
- Column %
- Chi-Square Test

Interpretation Required:

- Which group has higher productivity?
- Is there a statistically significant association?
- What does the chi-square result indicate?

Run another:

Food_Habit × Exercise_Level

Interpret the relationship.

Screen_Time_Group * Productivity_Level Crosstabulation					
			Productivity_Level		Total
			Low	High	
Screen_Time_Group	Low	Count	21	12	33
		% within Screen_Time_Group	63.6%	36.4%	100.0%
		% within Productivity_Level	28.8%	44.4%	33.0%
	Moderate	Count	15	8	23
		% within Screen_Time_Group	65.2%	34.8%	100.0%
		% within Productivity_Level	20.5%	29.6%	23.0%
	High	Count	37	7	44
		% within Screen_Time_Group	84.1%	15.9%	100.0%
		% within Productivity_Level	50.7%	25.9%	44.0%
Total		Count	73	27	100
		% within Screen_Time_Group	73.0%	27.0%	100.0%
		% within Productivity_Level	100.0%	100.0%	100.0%

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	4.921 ^a	2	.085
Likelihood Ratio	5.112	2	.078
Linear-by-Linear Association	4.210	1	.040
N of Valid Cases	100		
a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 6.21.			

Interpretation: Screen_Time_Group × Productivity_Level

The crosstab analysis shows the relationship between screen time group and productivity level among participants.

Participants with **low screen time** have the highest percentage of high productivity (**36.4%**), followed by the moderate screen time group (**34.8%**). In contrast, participants with high screen time show the lowest percentage of high productivity (**15.9%**).

This suggests that lower screen time may be associated with better productivity levels.

Chi-Square Test Interpretation

The Pearson Chi-Square test value is **4.921** with a significance value (**p-value**) of **0.085**.

Since the p-value is greater than **0.05**, there is **no statistically significant association** between screen time group and productivity level.

This means that although differences in productivity are visible across screen time groups, the relationship is not statistically strong enough to conclude that screen time significantly affects productivity in this dataset.

Food_Habit * Exercise_Level Crosstabulation						
			Exercise_Level			Total
			High	Low	Medium	
Food_Habit	Fast Food	Count	14	13	12	39

		% within Food_Habit	35.9%	33.3%	30.8%	100.0%
		% within Exercise_Level	43.8%	37.1%	36.4%	39.0%
		Healthy	Count	6	10	13
		% within Food_Habit	20.7%	34.5%	44.8%	100.0%
		% within Exercise_Level	18.8%	28.6%	39.4%	29.0%
		Moderate	Count	12	12	8
	% within Food_Habit		37.5%	37.5%	25.0%	100.0%
	% within Exercise_Level		37.5%	34.3%	24.2%	32.0%
	Total		Count	32	35	33
% within Food_Habit			32.0%	35.0%	33.0%	100.0%
% within Exercise_Level			100.0%	100.0%	100.0%	100.0%

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	3.646 ^a	4	.456
Likelihood Ratio	3.723	4	.445
N of Valid Cases	100		
a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 9.28.			

Interpretation: Food_Habit × Exercise_Level

The crosstab analysis shows the relationship between food habit and exercise level among participants.

Participants with healthy food habits have the highest percentage in the medium exercise level category (44.8%). On the other hand, participants with fast food habits show relatively higher percentages in the high exercise level category (35.9%).

This suggests that food habit may have some influence on exercise behavior among participants.

Chi-Square Test Interpretation

The Pearson Chi-Square test value is **3.646** with a significance value (p-value) of **0.456**.

Since the p-value is greater than **0.05**, there is no statistically significant association between food habit and exercise level.

This means that although some differences are observed between food habit groups and exercise levels, the relationship is not statistically strong enough to conclude that food habit significantly affects exercise level in this dataset.

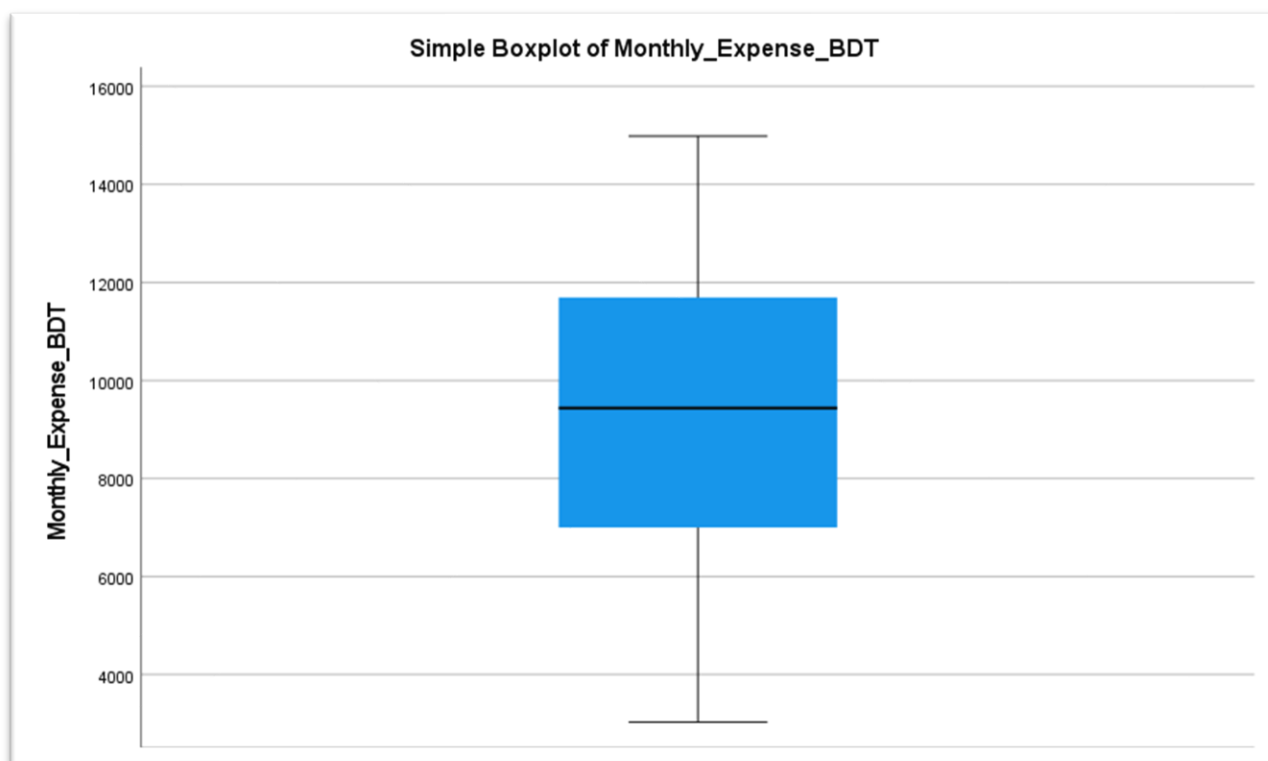
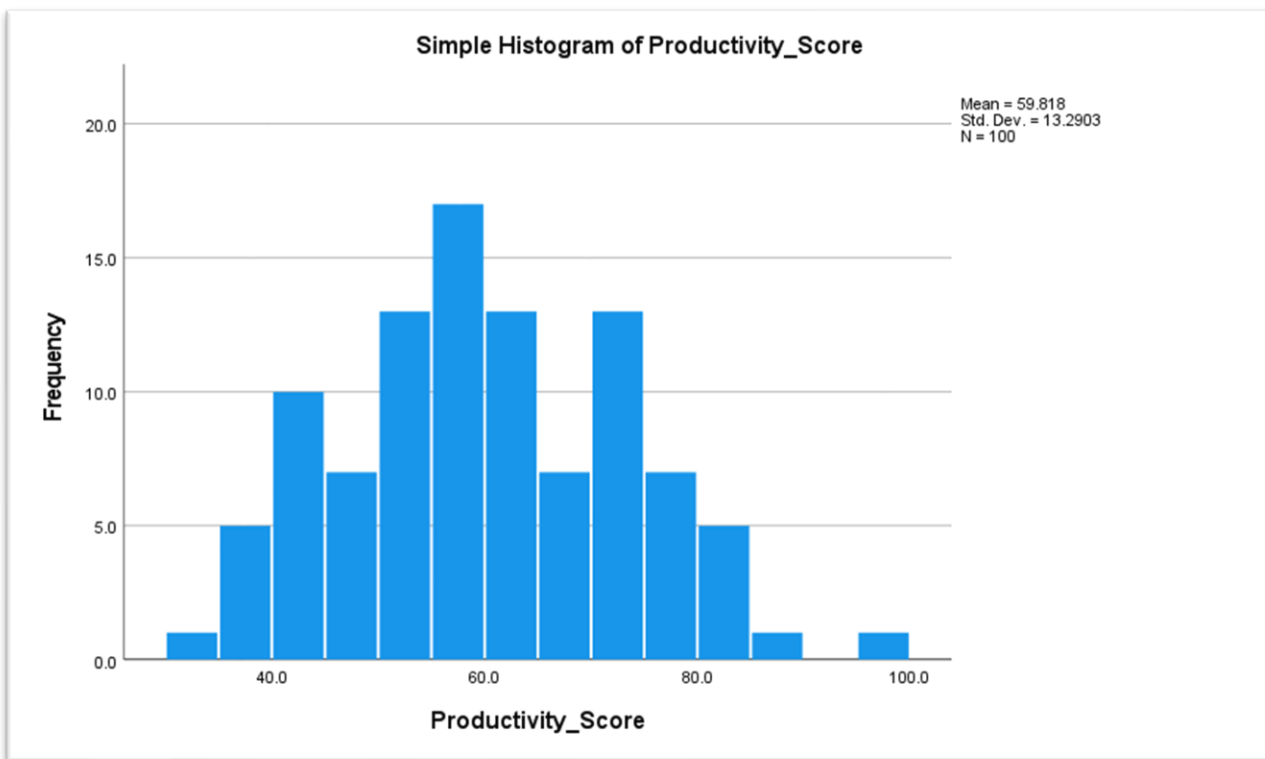
Task 4: Data Visualization (10 Marks)

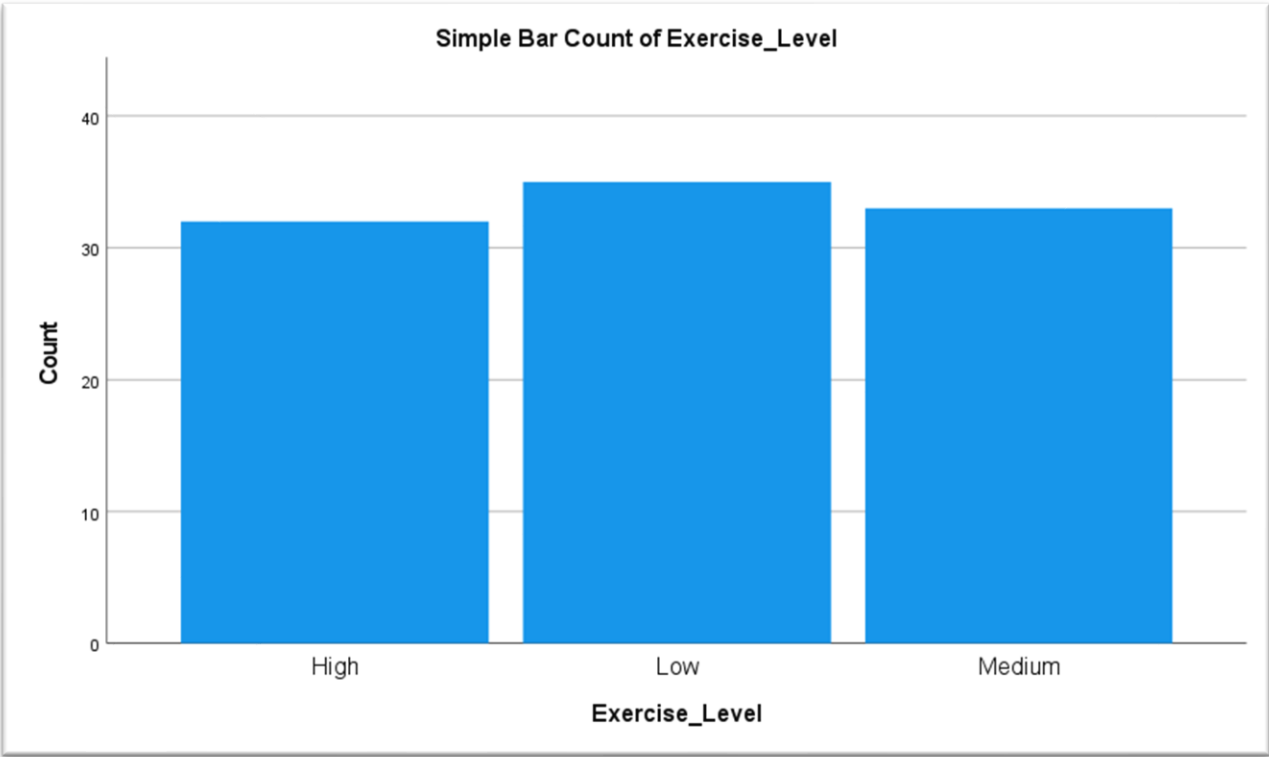
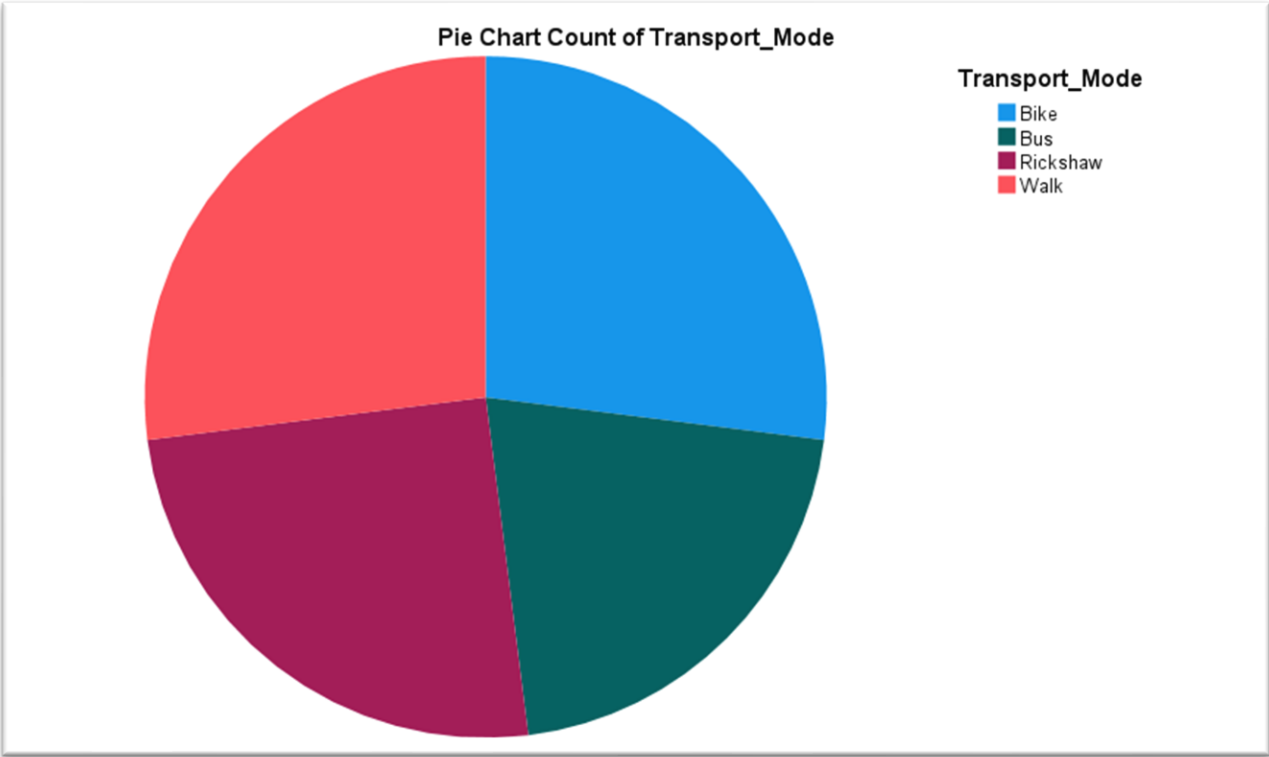
Create:

- Histogram → Productivity Score
- Boxplot → Monthly Expense
- Bar Chart → Exercise Level
- Pie Chart → Transport Mode

Interpretation Required:

- Describe the distribution pattern
- Identify any outliers
- Mention the most common categories
- Explain what each graph shows





The histogram shows the distribution of productivity scores among participants. The data appears approximately bell-shaped and symmetric, indicating a nearly normal distribution. Most participants scored between 50 and 70 in productivity.

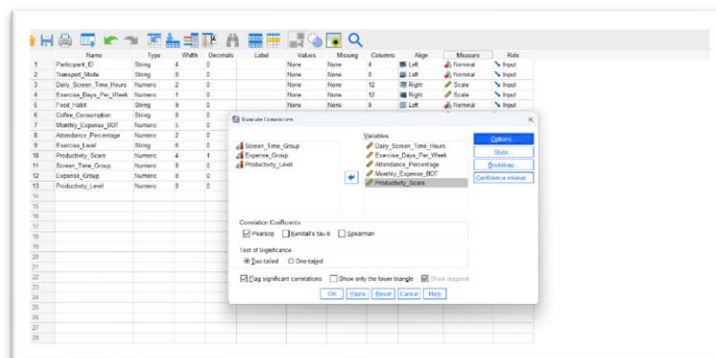
Boxplot Interpretation (Monthly Expense)

No significant outliers are visible because there are no extreme points outside the whiskers.

The bar chart represents the frequency of different exercise levels among participants. The “Low” exercise level has the highest frequency, making it the most common category. The frequencies of “High” and “Medium” exercise levels are relatively similar.

The pie chart shows the proportion of different transport modes used by participants. Bike and Walk appear to be the most commonly used transport modes, while Bus has the lowest proportion.

The chart helps visualize the percentage distribution of transport preferences among participants.



Correlations						
		Daily_Screen_ Time_Hours	Exercise_Day s_Per_Week	Attendance_ Percentage	Monthly_Ex pense_BDT	Productivi ty_Score
Daily_Screen_ Time_Hours	Pears on Correl ation	1	.046	.159	-.009	-.274**
	Sig. (2- tailed)		.652	.114	.930	.006
	N	100	100	100	100	100
Exercise_Days _Per_Week	Pears on Correl ation	.046	1	.162	.124	.781**
	Sig. (2- tailed)	.652		.107	.218	.000
	N	100	100	100	100	100
Attendance_Pe rcentage	Pears on Correl ation	.159	.162	1	.058	.391**
	Sig. (2- tailed)	.114	.107		.568	.000
	N	100	100	100	100	100
Monthly_Expe nse_BDT	Pears on Correl ation	-.009	.124	.058	1	.140

	Sig. (2-tailed)	.930	.218	.568		.165
	N	100	100	100	100	100
Productivity_Score	Pearson Correlation	-.274**	.781**	.391**	.140	1
	Sig. (2-tailed)	.006	.000	.000	.165	
	N	100	100	100	100	100
**. Correlation is significant at the 0.01 level (2-tailed).						

Daily Screen Time Hours & Productivity Score

There is a **negative weak correlation** between daily screen time and productivity score ($r = -0.274$). This means that as screen time increases, productivity tends to decrease.

The relationship is **statistically significant** because the p-value (0.006) is less than 0.05.

Practically, this suggests that spending more time on screens may reduce participants' productivity levels.

Exercise Days Per Week & Productivity Score

There is a **strong positive correlation** between exercise days per week and productivity score ($r = 0.781$). This indicates that participants who exercise more frequently tend to have higher productivity.

The relationship is **statistically significant** because the p-value is less than 0.001.

Practically, regular exercise may positively influence productivity levels.

Attendance Percentage & Productivity Score

There is a **moderate positive correlation** between attendance percentage and productivity score ($r = 0.391$). This means that higher attendance is associated with higher productivity.

The relationship is **statistically significant** because the p-value is less than 0.001.

Practically, participants with better attendance tend to be more productive.

Monthly Expense BDT & Productivity Score

There is a **weak positive correlation** between monthly expense and productivity score ($r = 0.140$). However, the relationship is very weak.

The relationship is **not statistically significant** because the p-value (0.165) is greater than 0.05.

Practically, monthly expense does not appear to have a meaningful effect on productivity in this dataset.

Task 6: Regression Analysis (15 Marks)

👉 Simple Linear Regression

Dependent Variable:

- Productivity_Score

Independent Variable:

- Daily_Screen_Time_Hours

Interpret:

- R^2
- Coefficient
- Significance value
- Overall model meaning

👉 Multiple Regression

Dependent Variable:

- Productivity_Score

Independent Variables:

- **Interpretation Required:**

- | Model Summary | | | | |
|---------------|-------------------|----------|-------------------|----------------------------|
| Model | R | R Square | Adjusted R Square | Std. Error of the Estimate |
| 1 | .274 ^a | .075 | .066 | 12.8452 |
- a. Predictors: (Constant), Daily_Screen_Time_Hours

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1316.452	1	1316.452	7.978	.006 ^b
	Residual	16170.015	98	165.000		
	Total	17486.468	99			

a. Dependent Variable: Productivity_Score

b. Predictors: (Constant), Daily Screen Time Hours

Coefficients ^a						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	69.076	3.520		19.623	.000
	Daily_Screen_Time_Hours	-1.241	.439	-.274	-2.825	.006
a. Dependent Variable: Productivity_Score						

Simple Linear Regression Interpretation

The simple linear regression analysis was conducted to examine the effect of daily screen time on productivity score.

The Model Summary table shows an **R² value of 0.075**, which means that daily screen time explains **7.5%** of the variation in productivity score.

The regression coefficient (B = -1.241) indicates a **negative effect** of screen time on productivity. This means that for every one-hour increase in daily screen time, the productivity score decreases by approximately **1.24 units**.

The ANOVA table shows a significance value (**p = 0.006**), which is less than **0.05**. Therefore, the overall regression model is statistically significant.

Overall, the results suggest that higher daily screen time significantly reduces productivity among participants.

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.900 ^a	.810	.802	5.9204
a. Predictors: (Constant), Monthly_Expense_BDT, Daily_Screen_Time_Hours, Exercise_Days_Per_Week, Attendance_Percentage				

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	14156.607	4	3539.152	100.971	.000 ^b
	Residual	3329.861	95	35.051		
	Total	17486.468	99			
a. Dependent Variable: Productivity_Score						
b. Predictors: (Constant), Monthly_Expense_BDT, Daily_Screen_Time_Hours, Exercise_Days_Per_Week, Attendance_Percentage						

Coefficients ^a						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	34.958	3.586		9.749	.000
	Daily_Screen_Time_Hours	-1.628	.205	-.360	-7.933	.000
	Exercise_Days_Per_Week	4.153	.256	.742	16.227	.000
	Attendance_Percentage	.293	.041	.327	7.110	.000
	Monthly_Expense_BDT	.000	.000	.026	.572	.569
a. Dependent Variable: Productivity_Score						

Multiple Regression Interpretation

The multiple regression analysis was conducted to examine the effects of daily screen time, exercise days, attendance percentage, and monthly expense on productivity score.

The Model Summary table shows an **R² value of 0.810**, which means that **81%** of the variation in productivity score is explained by the independent variables included in the model.

The ANOVA table shows a significance value (**p < 0.001**), indicating that the overall regression model is statistically significant.

Among all variables, **Exercise_Days_Per_Week** has the strongest effect on productivity because it has the highest standardized beta coefficient (**$\beta = 0.742$**). This indicates a strong positive relationship between exercise frequency and productivity.

Positive Effects

- **Exercise_Days_Per_Week** has a positive effect on productivity ($B = 4.153$). This means that increasing exercise days by one day increases productivity score by about 4.15 units.
- **Attendance_Percentage** also has a positive effect ($B = 0.293$), indicating that higher attendance is associated with higher productivity.

Negative Effect

- **Daily_Screen_Time_Hours** has a negative effect on productivity ($B = -1.628$). This means that increasing screen time by one hour decreases productivity score by approximately 1.63 units.

Weak/Non-significant Effect

- **Monthly_Expense_BDT** has a very weak positive effect and is not statistically significant because the p-value (0.569) is greater than 0.05.

Overall, the results suggest that regular exercise and better attendance improve productivity, while excessive screen time reduces productivity.

Task 7: Hypothesis Testing (15 Marks)

👉 Independent Sample t-test

Compare:

Productivity Score between:

- Male vs Female

✳️ **Interpretation Required:**

- Is there a significant mean difference?

- Explain the p-value
- State the final conclusion

👉 One-Way ANOVA

Compare:

Productivity Score among different Food Habits

📌 Interpretation Required:

- Is there any significant difference among groups?
- Which group appears to have the highest productivity?

👉 Chi-Square Test

Run:

Productivity_Level × Exercise_Level

📌 Interpretation Required:

- Is there an association between variables?
- Explain the significance result

Independent Sample T Test

Group Statistics					
	Gender	N	Mean	Std. Deviation	Std. Error Mean
Productivity_Score	Male	50	60.498	12.0397	1.7027
	Female	50	59.138	14.5248	2.0541

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
Productivity_Score	Equal variances assumed	2.243	.137	.510	98	.611	1.3600	2.6680	-3.9346	6.6546
	Equal variances not assumed			.510	94.741	.611	1.3600	2.6680	-3.9369	6.6569

The independent sample t-test was conducted to compare productivity scores between male and female participants.

The mean productivity score of males (**60.50**) is slightly higher than females (**59.14**).

The p-value is **0.611**, which is greater than **0.05**. Therefore, there is **no statistically significant mean difference** in productivity scores between male and female participants.

This means that gender does not have a significant effect on productivity score in this dataset.

Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)

Pearson Chi-Square	2.791 ^a	2	.248
Likelihood Ratio	2.913	2	.233
N of Valid Cases	100		
a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 8.64.			

Chi-Square Test Interpretation

The Chi-square test was conducted to examine the association between the variables.

The Pearson Chi-Square significance value (p-value) is **0.248**, which is greater than **0.05**.

Therefore, there is **no statistically significant association** between the variables.

This means that the relationship observed between the variables may have occurred by chance and is not statistically meaningful in this dataset.

★ Bonus (Optional – No Marks)

Try:

Logistic Regression

Dependent:

- Productivity_Level

Independent:

- Screen Time
- Exercise Days
- Attendance

✚ Interpret the odds ratio and significance.

Classification Table ^{a,b}			
	Observed	Predicted	
		Productivity_Level	
		Low	High
		Percentage Correct	

Step 0	Productivity_Level	Low	73	0	100.0
		High	27	0	.0
	Overall Percentage				73.0
a. Constant is included in the model.					
b. The cut value is .500					

Variables in the Equation							
		B	S.E.	Wald	df	Sig.	Exp(B)
Step 0	Constant	-.995	.225	19.499	1	.000	.370

Variables not in the Equation					
			Score	df	Sig.
Step 0	Variables	Daily_Screen_Time_Hours	3.255	1	.071
		Exercise_Days_Per_Week	40.532	1	.000
		Attendance_Percentage	13.363	1	.000
	Overall Statistics		54.027	3	.000

Omnibus Tests of Model Coefficients				
		Chi-square	df	Sig.
Step 1	Step	78.846	3	.000
	Block	78.846	3	.000
	Model	78.846	3	.000

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	37.806 ^a	.545	.792

a. Estimation terminated at iteration number 8 because parameter estimates changed by less than .001.

Classification Table^a

	Observed		Predicted		
			Productivity_Level		Percentage Correct
			Low	High	
Step 1	Productivity_Level	Low	68	5	93.2
		High	4	23	85.2
	Overall Percentage				91.0
a. The cut value is .500					

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1 ^a	Daily_Screen_Time_Hours	-.709	.242	8.605	1	.003	.492
	Exercise_Days_Per_Week	1.621	.401	16.324	1	.000	5.060
	Attendance_Percentage	.128	.041	9.570	1	.002	1.136
	Constant	-13.342	3.660	13.286	1	.000	.000
a. Variable(s) entered on step 1: Daily_Screen_Time_Hours, Exercise_Days_Per_Week, Attendance_Percentage.							

★ Logistic Regression Interpretation

Logistic regression was conducted to predict productivity level using daily screen time, exercise days per week, and attendance percentage.

The Omnibus Test shows a significance value of $p < 0.001$, indicating that the overall logistic regression model is statistically significant.

The Nagelkerke R^2 value is **0.792**, which means the model explains approximately **79.2%** of the variation in productivity level.

The classification table shows an overall prediction accuracy of **91%**, indicating that the model predicts productivity level quite well.

◆ Daily Screen Time Hours

The odds ratio $\text{Exp}(B)$ is **0.492** with a p-value of **0.003**.

Since $\text{Exp}(B)$ is less than 1, daily screen time has a **negative effect** on productivity level. This means that increasing screen time decreases the likelihood of having high productivity.

The relationship is statistically significant because the p-value is less than 0.05.

◆ Exercise Days Per Week

The odds ratio $\text{Exp}(B)$ is **5.060** with a p-value less than **0.001**.

This indicates a strong positive effect on productivity level. Participants who exercise more frequently are about **5 times more likely** to have high productivity.

The relationship is statistically significant.

◆ Attendance Percentage

The odds ratio $\text{Exp}(B)$ is **1.136** with a p-value of **0.002**.

This means that higher attendance increases the likelihood of high productivity. For every one-unit increase in attendance percentage, the odds of high productivity increase by about **13.6%**.

The relationship is statistically significant.